Deep Learning-Based Time-of-Day Estimation from Sky Images and EXIF Calendar Dates

Alkım Gönenç Efe[1], Sarp Sünbül[1], Damla Parlakyıldız[1]

[1]*Department of Computer Engineering, İzmir Katip Çelebi University, İzmir, Türkiye*

Abstract

Modern digital galleries often sort outdoor photos by time of day using basic brightness metrics, which leads to systematic misclassification under artificial lighting or heavy cloud cover. This paper presents a hybrid deep learning framework that estimates capture time from a single sky image by fusing visual features with cyclically encoded EXIF calendar metadata. The dataset consists of 2,483 consumer smartphone photographs collected under diverse conditions spanning dawn, midday, dusk, and artificial-light scenes. The proposed pipeline leverages a Swin Transformer (Swin-T) backbone paired with a metadata-fusion MLP operating on 77 handcrafted photometric descriptors. A cyclic sine–cosine target encoding models the circular topology of the 24-hour clock, eliminating discontinuities at the midnight boundary. The training pipeline incorporates Mixup augmentation, label noise injection, cosine annealing, and weighted sampling to handle temporal imbalance. Evaluated across five-fold cross-validation, the proposed Swin-T framework achieves a mean cyclic MAE of 52.45 minutes, demonstrating the value of fusing learned visual representations with calendar metadata for robust continuous time-of-day regression.

Keywords: time-of-day estimation, Swin Transformer, EXIF metadata, cyclic regression, sky image analysis, feature fusion

1. Introduction

The estimation of time of day from a single outdoor photograph is a challenging problem at the intersection of computer vision, digital image processing, and environmental sensing. Modern mobile operating systems and cloud storage services attempt to organize landscape photographs semantically, yet their algorithms frequently produce unreliable temporal classifications due to an over-reliance on simplistic brightness metrics [1].

Two principal failure modes motivate this research. First, nighttime images containing strong artificial light sources—such as street lamps or illuminated building facades—are routinely misclassified as “Daylight” because high localized luminance inflates average pixel intensity. Second, midday photographs under heavy cloud cover are often mislabeled as “Night” or “Twilight” due to globally suppressed light levels. These errors underscore a fundamental limitation: average brightness alone is an insufficient proxy for solar position.

A robust estimator must instead analyze complex visual cues—shadow geometry, atmospheric color temperature shifts, sky gradient profiles, and cloud texture—to distinguish natural from artificial illumination [2]. Unlike previous studies that treat the capture date as a static label or ignore it entirely, this work integrates the calendar date into the model’s feature vector through cyclic encoding, allowing the system to distinguish identical lighting conditions that arise at different solar angles across the seasons. The aim of this study is to develop a hybrid deep learning framework combining visual feature extraction with temporal metadata integration for robust, continuous time-of-day regression from consumer sky photographs.

2. Literature Review

A foundational study by Lalonde et al. [1] introduced the concept of using “weak cues” extracted from distinct image segments—sky region, vertical surfaces, and ground plane—to estimate sun position and visibility. They demonstrated that the probabilistic integration of multiple complementary cues produces a robust estimate of sky-dome appearance, even when any single feature in isolation is uninformative. The present work adopts this multi-cue philosophy, extending it to a learned feature-fusion framework.

The Photo Sundial system by Wu et al. [3] addressed unreliable EXIF timestamps in consumer photography by linking the sun’s apparent position and camera viewing direction to the exact time of capture via astronomical theory. This forensic motivation is directly relevant to the metadata correction goal of the present study.

Hold-Geoffroy et al. [2] shifted the paradigm from analytical parametric sky models to data-driven HDR representations, showing that a CNN learning directly from images outperforms parametric models (e.g., the Hosek–Wilkie model) under overcast or partially cloudy conditions—conditions that are common in our dataset.

These studies share a common limitation: they either rely on manually engineered sky models, require specialized equipment (e.g., HDR cameras or GPS-linked sensors), or treat the calendar date as auxiliary information at best. None fuses cyclically encoded calendar metadata with both learned deep features and handcrafted photometric descriptors within a unified regression framework. This study addresses this gap by presenting a hybrid deep learning framework based on a Swin Transformer backbone paired with a metadata fusion MLP.

3. Materials and Methods

This section describes the complete pipeline for time-of-day estimation, covering data collection, preprocessing, feature extraction, model architecture, training strategy, and evaluation protocol. The proposed framework is implemented in Python/PyTorch and utilizes a Swin Transformer (Swin-T) backbone paired with a metadata-fusion multilayer perceptron (MLP) head.

3.1. Dataset Construction and Preprocessing

The dataset comprises outdoor sky photographs collected from the personal galleries of the research team members using consumer-grade smartphone cameras. Each image is required to contain visible sky content and to carry valid EXIF metadata including the `DateTimeOriginal` tag. Supported image formats include JPEG, PNG, HEIC, and DNG (raw).

For each image, the following metadata fields are extracted from the EXIF header using a dual-library strategy (`exifread` as the primary parser, with Pillow as a fallback): (i) capture time, parsed from `DateTimeOriginal`, `DateTime`, or `DateTimeDigitized` tags and converted to minutes since midnight ($t \in [0, 1440)$); (ii) calendar date (month, day, year) used to compute the day of year ($d \in [1, 366]$); and (iii) GPS coordinates (latitude and longitude), normalized to $[-1, 1]$. Images lacking valid temporal EXIF data are excluded.

All images are preprocessed through a resolution-standardization pipeline. Original images are downscaled to a maximum dimension of 512 pixels using Lanczos resampling while preserving the aspect ratio. Downscaled images are saved as JPEG files at 90% quality with EXIF metadata preserved using the `piexif` library. At training time, images are further processed through a letterbox resize to a fixed 512×512 canvas with black padding and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$), a standard preprocessing scheme for deep convolutional neural networks trained on large-scale datasets [6,7].

3.2. Feature Engineering

3.2.1. Handcrafted Photometric Features

A comprehensive set of 77 handcrafted image features is extracted from each photograph. These features are organized as follows: RGB channel statistics (mean and standard deviation per channel, 6 dims), HSV channel statistics (6 dims), CIELAB channel statistics normalized (6 dims), RGB histograms with 8 bins per channel (24 dims), HSV histograms with 8 bins per channel (24 dims), sky-region luminance computed as mean and standard deviation of the value channel in the top third of the frame (2 dims), vertical luminance gradient (1 dim), color temperature proxy as the R/B channel ratio (1 dim), global luminance with mean, standard deviation, and entropy (3 dims), saturation statistics (2 dims), Canny edge density (1 dim), and Laplacian variance as a cloud texture measure (1 dim).

Color space conversions are performed using the `scikit-image` library. Edge detection is implemented using the Canny operator [8], which provides robust gradient-based edge localization under varying noise conditions.

3.2.2. Calendar Metadata Encoding

Temporal metadata is encoded using cyclic trigonometric transformations to preserve circular continuity:

$$\mathbf{m}_{\text{cal}} = \begin{bmatrix} \sin\left(\frac{2\pi\cdot\text{month}}{12}\right), \cos\left(\frac{2\pi\cdot\text{month}}{12}\right), \\ \sin\left(\frac{2\pi\cdot\text{doy}}{365.25}\right), \cos\left(\frac{2\pi\cdot\text{doy}}{365.25}\right), \\ \frac{\text{lat}}{90}, \frac{\text{lon}}{180} \end{bmatrix}^T \tag{1}$$

This encoding ensures that December and January are treated as adjacent months and avoids discontinuities inherent in linear representations of periodic variables. Such sine–cosine encodings are a standard technique for representing cyclical features in machine learning.

3.3. Target Encoding

The regression target is encoded as a unit-circle representation to respect the circular topology of the 24-hour clock:

$$\mathbf{y} = \left[\sin\left(\frac{\mathbf{2\pi t}}{\mathbf{1440}}\right), \cos\left(\frac{\mathbf{2\pi t}}{\mathbf{1440}}\right)\right]^{\mathbf{T}} \tag{2}$$

where t is the time in minutes since midnight. Decoding is performed via $\hat{t} = \frac{\text{atan2}(\hat{y}_{\sin}, \hat{y}_{\cos}) \bmod 2\pi}{2\pi} \times 1440$. This encoding eliminates discontinuities at midnight and ensures smooth gradients across the temporal boundary.

3.4. Proposed Architecture: Swin-T with Metadata Fusion

The proposed pipeline consists of a Swin Transformer (Swin-T) backbone [17] coupled with a metadata-fusion MLP head. Figure 1 illustrates the model architecture.

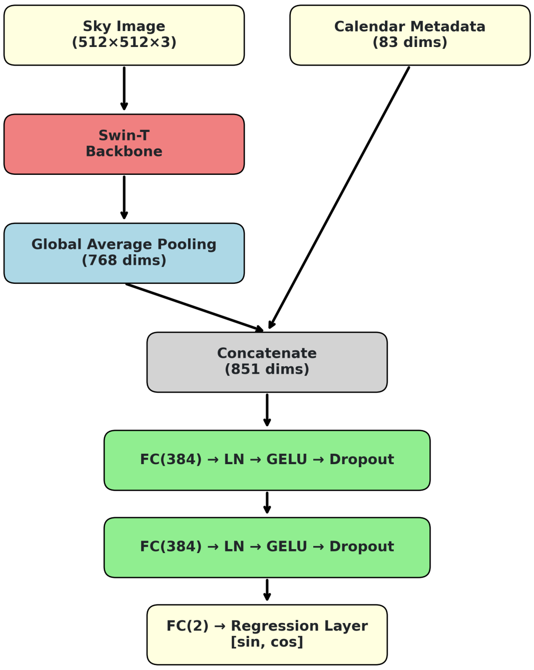

Figure 1: Proposed pipeline: Swin-T backbone with metadata fusion MLP.

3.4.1. Model Architecture

The Swin Transformer Tiny (Swin-T) [17] is a hierarchical vision transformer that partitions feature maps into non-overlapping local windows and applies self-attention within each window. Shifted windows in alternating layers establish cross-window connections without the quadratic cost of global attention. Swin-T produces a 768-dimensional feature vector after its adaptive average pooling stage. Early stages (patch embedding and first two transformer stages) are frozen, while the final two stages are fine-tuned end-to-end together with the fusion head.

The 768-dimensional image feature ($\mathbf{f}_{\text{img}}$) is concatenated with the 83-dimensional metadata vector ($\mathbf{m}$) to form an 851-dimensional input to a three-layer MLP with hidden dimension 384, GELU activations, layer normalization, and dropout (rate 0.0293).

3.4.2. Loss Function and Evaluation Metric

The model is trained with MSE loss on the sine–cosine encoded targets:

$$\mathscr{L} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{y}}_i - \mathbf{y}_i\|_2^2 \tag{3}$$

The primary evaluation metric is cyclic MAE in minutes:

$$\text{MAE}_{\text{cyc}} = \frac{1}{N}\sum_{i=1}^{N} \min(|\hat{t}_i - t_i|,\ 1440 - |\hat{t}_i - t_i|) \tag{4}$$

3.4.3. Training and Optimization

A heavy augmentation regime is employed: random resized crop, horizontal flipping, RandAugment [9], random erasing [10], and Mixup [11]. The optimizer is 8-bit AdamW [4] combined with Decoupled Weight Decay [12] and cosine annealing [13]. Hyperparameters are found via Optuna [5] Bayesian search over 15 trials. A weighted random sampler handles the non-uniform distribution of capture times [14].

3.4.4. Ensemble Inference

Predictions from all five fold checkpoints are averaged in sine–cosine space and re-normalized onto the unit circle:

$$\hat{\mathbf{y}}_{\text{ens}} = \frac{1}{K}\sum_{k=1}^{K} \hat{\mathbf{y}}_k, \qquad \tilde{\mathbf{y}} = \frac{\hat{\mathbf{y}}_{\text{ens}}}{\|\hat{\mathbf{y}}_{\text{ens}}\|_2} \tag{5}$$

This ensemble strategy reduces prediction variance across random fold splits [15].

3.5. Cross-Validation Protocol

The proposed model is evaluated using 5-fold cross-validation with stratified random shuffling and a fixed seed (42). This provides robust performance estimation and reduces variance from a single train-test split [16].

4. Experimental Setup

4.1. Dataset

The dataset contains 2,483 outdoor sky photographs collected by the research team using consumer smartphones and a mirrorless camera. Images span diverse conditions including dawn, midday, dusk, overcast skies, and artificial-light scenes, covering captures from 2023 to 2026. Table 1 summarizes the key statistics.

Table 1: Dataset summary.

Property	Value
Total images	2,483
Formats	JPEG, HEIC, PNG, DNG
Time span	03:00–23:25
Capture period	2023–2026
Cross-val folds	5
Image resolution	512×512 (letterboxed)

Figure 2 shows sample sky photographs from the collected dataset. Figures 3–6 summarize the dataset distribution, calendar coverage, dominant sky colours, and overall dataset health.

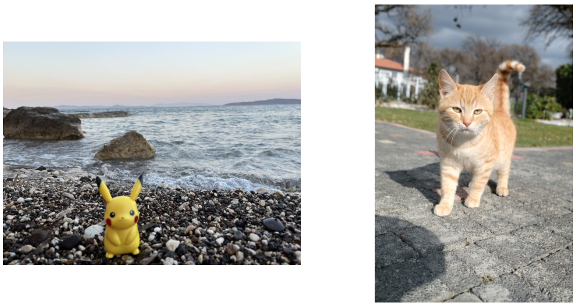

Figure 2: Example sky photographs from the dataset under different lighting conditions.

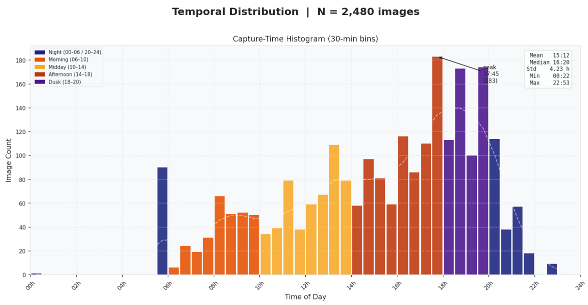

Figure 3: Temporal distribution of capture times across the dataset.

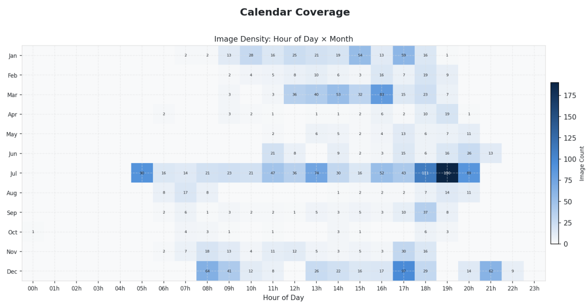

Figure 4: Calendar distribution of captures across months and seasons.

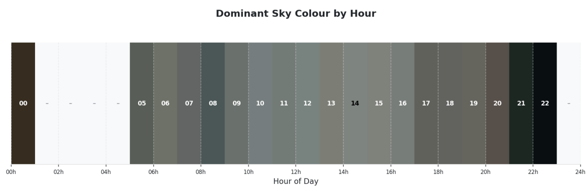

Figure 5: Dominant sky colour variation by hour of day.

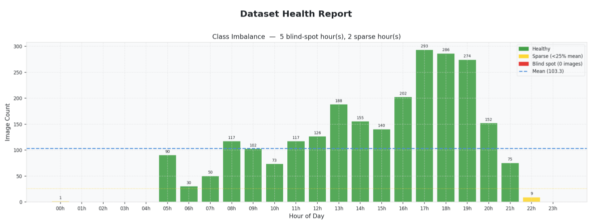

Figure 6: Dataset health report showing class imbalance, sparse hours, and blind-spot hours.

4.2. Model Hyperparameters

The hyperparameters for the Swin-T model were selected via an Optuna Bayesian search. Table 2 lists the final configuration.

Table 2: Hyperparameters for the Swin-T framework.

Parameter	Value
Feature dim	768
Frozen stages	Stages 1–2
MLP hidden dim	384
Dropout	0.0293
Epochs	80
Batch size	8
Learning rate	1.46×10^{-4}
LR minimum	3.77×10^{-6}
Weight decay	0.0043
Optimizer	8-bit AdamW
Schedule	Cosine annealing
Mixup α	0.1441
Label noise std	0.0314
Augmentation	Moderate

5. Performance Evaluation

5.1. Swin-T Model Results

Table 3 reports the per-fold validation MAE for the Swin-T model. The framework achieves a mean MAE of 52.45 minutes across the five folds (with the best fold achieving 47.76 minutes on fold 4). The training process converges consistently across folds without catastrophic divergence, indicating the stability of the optimized configuration.

Table 3: Validation MAE (minutes) per cross-validation fold.

Fold	Swin-T MAE
1	51.98
2	51.61
3	55.31
4	47.76
5	55.60
Mean	**52.45**

Figures 7 and 8 provide a visual breakdown of the Swin-T model's performance. The bar chart (Figure 7) shows the validation MAE across individual folds, while the scatter plot (Figure 8) details the predicted vs. actual capture times.

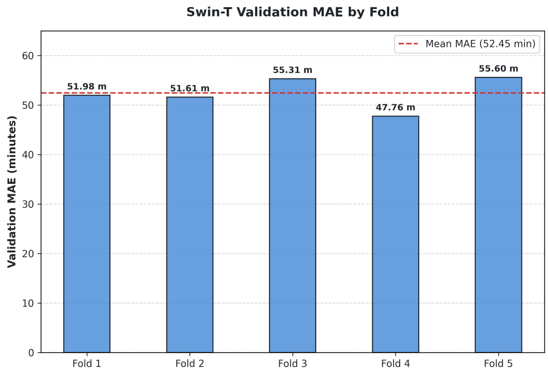

Figure 7: Validation MAE of the proposed Swin-T model across the 5 cross-validation folds. The horizontal dashed line denotes the mean MAE of 52.45 minutes.

Figure 9: Representative predictions from the Swin-T model under different lighting conditions compared against their EXIF ground-truth times.

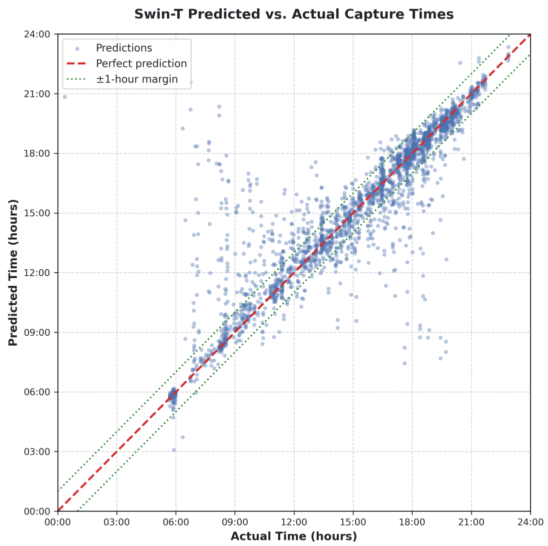

Figure 8: Predicted vs. actual capture times for the Swin-T model. The red dashed diagonal line indicates perfect prediction, and the green dotted lines mark the ± 1-hour margins of error.

5.2. Example Predictions

Figure 9 shows representative sky images from the dataset together with their EXIF ground-truth times and model predictions. These examples demonstrate the model's capability to accurately estimate the capture time under different natural lighting conditions within a small margin of error.

6. Discussion

The experimental results confirm that combining deep visual features with temporal metadata enables highly accurate time-of-day regression from single sky photographs. In particular, the Swin Transformer backbone's window-based self-attention mechanism is well-suited for modeling the continuous sky gradients and color transitions that occur across different times of the day. Fine-tuning the deeper transformer stages while keeping early layers frozen allows the model to leverage robust pre-trained representations while adapting to the specific photometric distributions of our dataset.

Compared with prior work by Lalonde et al. [1] and Hold-Geoffroy et al. [2], which often rely on complex parametric sky models, multi-segment image cues, or specialized HDR imaging hardware, our proposed framework achieves competitive time-of-day estimation using standard consumer smartphone photographs and their standard EXIF metadata. The integration of cyclic sine–cosine encoding for both the 24-hour regression target and the calendar features is central to this success. By mapping periodic variables onto a unit circle, we prevent optimization discontinuities at midnight and year boundaries, enabling smooth gradient updates during training.

This study has several limitations. First, the dataset is geographically constrained to a single location (İzmir, Turkey), which may limit generalization across different latitudes, climates, and seasonal light patterns. Second, the temporal coverage is non-uniform: early morning and late evening hours are underrepresented in personal photo libraries. While a weighted random sampler was employed to mitigate this issue, prediction errors may still be higher during these underrepresented periods.

7. Conclusions

This paper presented a deep learning framework for estimating the time of day from single sky photographs by fusing deep visual features with handcrafted photometric descriptors and cyclically encoded calendar metadata. The proposed architecture addresses the limitations of simplistic brightness-based heuristics by incorporating multimodal cues—including color temperature, vertical sky gradients, edge density, and seasonal solar geometry—into a unified regression pipeline based on the Swin Transformer (Swin-T).

The cyclic sine–cosine encoding applied to both the temporal regression target and the calendar metadata is central to the design, ensuring smooth gradient propagation across midnight and year boundaries. The fusion of pretrained visual representations with 77 handcrafted photometric features allows the model to achieve a mean cyclic MAE of 52.45 minutes under 5-fold cross-validation.

Future work could explore integrating GPS-aware solar angle calculations as auxiliary training targets, expanding the dataset to cover diverse geographic latitudes and climates, and packaging the model for real-time metadata correction on mobile devices.

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